

Literature review on Semantic Segmentation applied to martian terrain navigation datasets

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abstract This report presents a literature review on semantic segmentation tools and its application classifying images from the MSL and M2020 martian missions for the purpose of navigation. The first section explores the definition and basic structure of some widely used semantic segmentation models, with focus on the DeepLabV3+ model, and it also presents the difference between pixel-level and region-level loss functions. The second section explores the effectiveness of such models on performing semantic segmentation on the AI4Mars and similar independent datasets, presenting the main challenges to be yet solved in the efforts for a fully autonomous martian rover in the future.

I. SEMANTIC SEGMENTATION & RELATED MODELS

Definitions

The works by Ghosh et Al [?], Minaee et Al [1], Sultana et Al [2] and Kale et Al [?] explore the evolution and different possible deep learning techniques for image segmentation.

Image segmentation is an image processing technique used to divide an image into two or more meaningful regions, or alternatively, the process of defining boundaries separating semantic entities in an image. In a more technical definition, the task of image segmentation is to assign a label to each pixel within an image, with each label connected with respect to some property [3].

Image segmentation can be divided in three categories: Instance segmentation, in which different instances of the same semantic label are categorized independently; semantic segmentation, which assigns different labels for regions with different semantic meaning without classifying independent instances independently; and panoptic segmentation, which performs both tasks simultaneously [2].

The task of image segmentation has been significantly impacted by the development and advancement of deep learning techniques in computer vision, in particular after the success of AlexNet in 2012 in using CNN for Semantic Segmentation. Since then, numerous Deep Learning models for semantic segmentation have been developed with significant success [2], [4], [1].

Convolutional Neural Networks and Residual Networks

Classification tasks over an image require as output a probability distribution over a (at least) 2D image. Traditional fully connected networks (perceptrons) are unable

to capture spatial relations due to the linearization requirements. Convolutional networks don't suffer from this issue, since convolutions can be trivially extended over multiple channels it's possible to input images in the neural network while retaining the spatial relations at all scales. Convolutional networks use the convolution operation mixed with layers that resize images to try to extract information on these multiple scales [3].

Convolutional layers are the basis for a huge class of models for semantic segmentation include Convolutional Neural Networks (CNNs), the define boundary between regions by apply kernel-based convolutions to capture the positional relations of features in images. By adding pooling and unpooling layers that change the original size, it allows CNNs to find meaningful information at multiple scales of the feature image by following a "pyramid-like" design. CNNs are among the most successful approaches to semantic segmentation [1]. Spatial Pyramid Pooling is a pooling strategy that resizes/pools the image from larger to smaller to extract information on all scales.

A relevant advance in convolutional neural networks is the idea of dilated (or "atrous") convolutions [1], [5], which add an additional parameter to convolutions that increases the perceptive field without adding new parameters, according to the dilation rate r , shown in equation 1, where x represents the input and w represents the kernel. A visual representation of the effects of the dilation rate in a 3×3 kernel is shown in figure 1. In practice, the dilation rate makes the convolution pick samples farther from the center of the kernel, "skipping" neighbors, this increases the field of view without increasing the Kernel size.

$$y_i = \sum_{k=1}^K x[i + rk]w[k] \quad (1)$$

A work by He et Al [6] in 2016 proposed a new strategy to build deep neural networks without multiple layers while avoiding the problem of the vanishing gradient, notably, the degradation caused by increasing the number of layers isn't an over fitting problem. The paper addresses the gradient degradation by suggesting a Residual Block as part of the Neural Network structure. In a residual block, the input of an earlier layer is added to the the output of a deeper layer. So instead of an usual convolutional layer, where the output of a block of layers can be written as output = $F(x)$, a residual

block has an output of $F(x) + x$, as shown in figure 2. The work by He et al also shows that residual blocks allow for deeper networks to still improve and learn with less risk of gradient degradation, training networks with up to 152 layers and still seeing meaningful improvements in relation to smaller networks.

The basis for the DeepLab family of models are a combination of diluted convolutions for spatial pooling (instead of non-learning static pooling layers) with residual blocks, to extract/encode the information of the original input into a multi channel output, and a CNN for decoding the encoded input and generate the predictions for classification. This combination of convolution layers and residual blocks to extract the features of the original image in multiple scales and stack them together in a multichannel output. In traditional pyramid networks, this final layer is already close to the output as shown in figure 3. The DeepLab model adds complexity by using the pyramid-like section of it's design to just encode the information of the original input, the encoded features are then extracted by another CNN structure from the multi-channel pyramid output, as represented in figure 4.

Loss Functions

Another important choice in implementing a deep learning model is the choice of the loss function, a metric that evaluates how far a given output is from the expected ground truth. A good choice of loss function can mitigate usual challenges of image vision tasks, like noisy datasets and unbalanced class representation. A number of loss functions are presented and evaluated by the work done in [7]. Pixel-level loss functions utilize the information in all pixels of the prediction to score the model they usually result in increased accuracy, specially in scenarios where fine-grained accuracy and recall are necessary.

One of the most common pixel-level loss metrics is the Cross-entropy (CE) loss function. It measures the difference between two probability distributions given a random variable. In segmentation tasks, this means comparing the ground truth with the assigned probability that the pixel is of that class according to the model (i.e. output of the soft max function). The formula for the cross entropy is based on Shannon's entropy, and is presented in a practical formulation in equation 2, t_n is a one-hot encoded vector that is 1 on the target class and zero on the others, and $\hat{y}$ is the predicted probability. In imbalanced datasets, one

common approach to minimize the effects of the imbalance in the training of the model is adding weights that favor the less represented classes. If w is a vector of the weights of each labeled class, the weighted cross entropy is presented in equation 3.

$$L_{CE} = - \sum_{i=1}^C \log(t_i \cdot \hat{y}_i) \quad (2)$$

$$L_{CE} = - \sum_{i=1}^C t_i \cdot w \log(t_i \cdot \hat{y}_i) \quad (3)$$

Cross-entropy loss is part of the Pixel level loss functions, this family of loss functions evaluate each individual pixel as a part of their computation for dissimilarity, and they usually improve pixel-based assessment. Another approach are region-level loss functions, that focus on scoring based on the capacity of a model to adequately separate different semantic regions of an image. One example of region-level loss function is the Dice Loss function, that scores similarity based on the intersection between the predicted subset of pixels Y and the ground truth region T , and is written as in equation 4 for two class classification task, where $|A|$ represents the size of a set A . Form multi class segmentation tasks, the formula is presented in equation 5, where $\hat{y}$ represent the prediction probability of each class, and t is the vector that represents the ground truth. There is a choice in the number of classes considered in the context of a possible background/negative class, this background class is sometimes excluded from loss computations.

$$L_{Dice} = 1 - \frac{2|Y \cap T|}{|Y| + |T|} \quad (4)$$

$$L_{Dice} = 1 - \frac{1}{C} \sum_{c=1}^C \frac{2 \sum_i t_{ic} \hat{y}_{ic}}{\sum_i t_{ic} + \sum_i \hat{y}_{ic}} \quad (5)$$

A variation of the Dice Loss that offer some advantages in segmentation tasks is the log cosh Dice Loss, shown in equation 6, it enhances smoothness and robustness to outliers, and minimizes the effects of outliers, and it's a better behaved function in terms of derivatives, that helps the

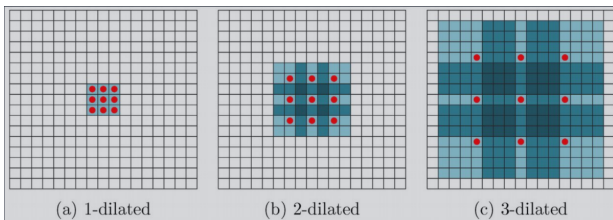


Fig. 1: Visual representation of diluted convolution with a 3x3 Kernel at different dilution rates. [1]

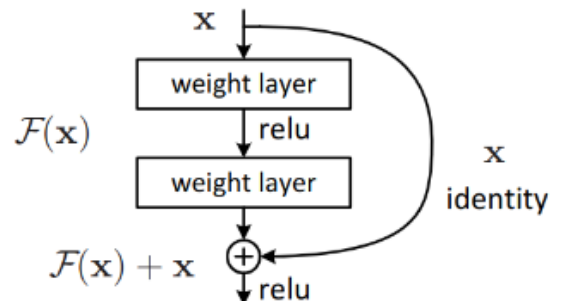


Fig. 2: Residual Block [6]

optimization steps. The Dice Loss can also be “weighted” in a sense. A generalized version of the dice loss is defined by the Generalized Wasserstein Dice Loss, the full scope of the generalization is outside of this work scope, but a less general version of weighted dice loss was used in one of the training steps, presented in subsection ??.

$$L_{log-cosh} = \log(\cosh(L_{Dice})) \quad (6)$$

II. MARTIAN TERRAIN SEGMENTATION RELATED WORK

The task of using computer vision for autonomous driving capability has been explored by NASA since 2004 [8] with the AutoNav project. The advancements in computer vision through deep learning models presented a great opportunity to improve the autonomous driving capabilities of the two active Martian rovers: Curiosity, launched as part of the Mars Science Laboratory (MSL) mission, and Perseverance, launched as part of the Mars 2020 (m2020) mission. Since advancements in computer vision fueled by development of deep-learning techniques, several research groups have attempted to use pictures from martian images, both from rovers and satellites, as input in segmentation tasks, with goal of identifying meaningful geological features or terrain for navigation. [8], [9], [10], [11]. The AI4Mars dataset is just of the datasets explored by this groups, while others have independent dataset, or have labeled the available images through different methods, or even implemented some level of semi-supervised learning algorithms [10].

The AI4Mars navigational dataset divides martian terrain in four classes [8] :

- **Soil:** Good for navigation
- **Bedrock:** Ideal for navigation.
- **Sand:** A less favorable but still viable path choice.
- **Big Rock:** Must be avoided while choosing a path.

The AI4Mars dataset is a publicly available set of images and semantic segmentation masks developed through a citizen science initiative by NASA. A number of volunteers were trained to create the masks, and their work was reviewed by specialist laborers and condensed into the final dataset [8]. One notable challenge in the dataset is that the Big Rock class is underrepresented, consisting of less than 2% of pixels in the m2020 dataset, as shown in Figure ??.

The work by Swan et al [8] trained a DeepLabV3+ with ResNet-101 on two versions of the MSL dataset, created by

two different labeling strategies called *random*, that used one possible mask from multiple suggested by trained labeler, and the *merged* strategy, where the final masks was a merged version of all suggestion by trained labelers, compiled by an expert on martian terrain. Their work showed that the model performed significantly better with the *merged* masks, with the accuracy for big rocks increasing from 82.83% to 93.24%. The final confusion matrices from this work are presented in table I

Actual	Predicted			
	Soil	Bedrock	Sand	Big Rock
<i>MSL Random (Overall Accuracy: 94.97%)</i>				
Soil	96.00	0.31	3.69	0.00
Bedrock	6.15	90.87	2.54	0.44
Sand	0.25	3.23	96.51	0.01
Big Rock	11.67	0.03	5.48	82.83
<i>MSL Merged (Overall Accuracy: 96.67%)</i>				
Soil	99.10	0.32	0.57	0.01
Bedrock	3.64	94.90	0.37	1.09
Sand	0.88	5.62	93.45	0.05
Big Rock	6.76	0.00	0.00	93.24

TABLE I: MSL NAVCAM Confusion Matrix Percentages for *random* and *merged* labeling strategies from Swan et al [8]

The work by Atha et Al [9] explores additional machine learning models for semantic segmentation of martian terrain in the context navigation, exploring how different models perform both on the MSL and on the M2020 navigational dataset. Their work identified two main problems with current models that aim to allow for independent path making for martian rovers. First, models tend to perform worse as the rovers move, and terrain around them changes naturally, the current models aren’t general enough for the martian surface. Second, the accuracy necessary for a model to perform independently, that is, without a remote human operator offering input, is significantly higher than the accuracy obtained by models at the time of that work.

For their model, they trained a ResNet-101 from zero with both MSL and M2020 datasets, they also trained a ResNet-101 from the weights for ImageNet, and they also tried to transfer train the model trained on the larger MSL and MER datasets to the M2020 dataset. The transfer learning did improve performance over soil and rock, but decreased

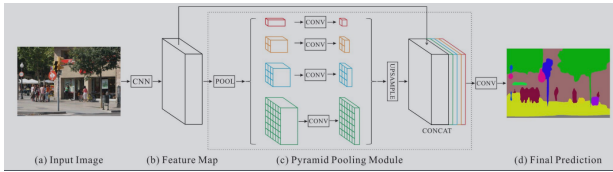


Fig. 3: The PSPNet architecture. A CNN produces the feature map and a pyramid pooling module aggregates the different sub-region representations. Up-sampling and concatenation are used to form the final feature representation from which, the final pixel-wise prediction is obtained through convolution [1].

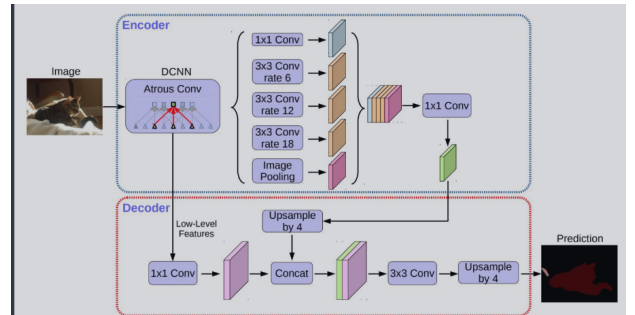


Fig. 4: The DeepLabV3+ model. From [1].

Metric	Class	ResNet-101	+ImageNet	+MSL
Precision	Soil	0.933	0.999	0.916
	Bedrock	0.857	0.467	0.917
	Sand	0.664	0.667	0.754
	Bigrock	0.558	0.618	0.174
Recall	Soil	0.746	0.596	0.838
	Bedrock	0.570	0.958	0.536
	Sand	0.975	0.952	0.957
	Bigrock	0.283	0.355	0.640

TABLE II: Performance metrics for models trained on the M2020 dataset by Atha et al. [9], reorganized for single-column fit.

accuracy on labeling bedrock. Their results in relation to the m2020 Dataset are presented in table II.

Luna et al [11] implemented ViBEKO’s DeepLabV3+ as part of their efforts to develop FASTNAV, their tool for independent rover navigation. The dataset used was independently obtained and labeled by them. Their implementation of DeepLabV3+ with ResNet-50 as a backbone achieved 87% accuracy and 72% mIoU with a time to prediction of 25ms, relevant for practical use in a rover.

Another interesting work in an independent dataset is the work of Goutham et al [10] that attempted to apply self-supervised methods to classify martian terrain using transformers on the AI4Mars dataset. As a first step, they trained a K-means clustering algorithm to explore what a deep learning model “would see” in relation to texture, structure and specific features of the images in the dataset. They also implemented a convex hull method for similar purposes. In relation to the dataset, they made significant effort into pre-processing the images and removing any information that wasn’t relevant for the classification task, such as covering the sky and rover in black pixels. They then trained different versions of the SegFormer based transformer, and obtained mIoU of more than 80% in all their models, with accuracy close to 90%.

III. CONCLUSION

Semantic segmentation is an elaborate and non-trivial task in computer vision, and performing segmentation on usual terrain such as martian terrain proves a challenge even for state of the art models. The lack of a very large dataset, and that the planet Mars itself offers a variety of very diverse terrain also limits the durability of successful models, a highly accurate model for the current region of a rover might perform significantly worse after the rover traveled enough for it’s immediate terrain to change significantly. This is also reinforced by the fact that transfer learning from the MSL to the M2020 dataset doesn’t seem to be particularly effective. Fully autonomous navigation would be a great advancement for interplanetary exploration on Mars, and modern deep learning models have made great stride in providing very useful navigational tools, but even better and more robust models are necessary for fully autonomous navigation in martian images.

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